



Applying Roots and Powers

You can apply roots and powers to many real-world situations.

Measurements may involve roots and powers.

Applications with Square Roots

EXAMPLE 1

What is the perimeter of a rectangle with a length of $6 + \sqrt{2}$ units and a width of $2 + \sqrt{6}$ units?

SOLUTION

perimeter = length + length + width + width

$$\begin{aligned} &= (6 + \sqrt{2}) + (6 + \sqrt{2}) + (2 + \sqrt{6}) + (2 + \sqrt{6}) \\ &= 6 + 6 + 2 + 2 + \sqrt{2} + \sqrt{2} + \sqrt{6} + \sqrt{6} \\ &= 16 + 2\sqrt{2} + 2\sqrt{6} \end{aligned}$$

The perimeter is $16 + 2\sqrt{2} + 2\sqrt{6}$ units. ■

EXAMPLE 2

What is the area of a rectangle with a length of $4\sqrt{2}$ units and a width of $5 + \sqrt{6}$ units?

SOLUTION

$$\begin{aligned}\text{area} &= \text{length} \cdot \text{width} \\ &= (4\sqrt{2})(5 + \sqrt{6}) \\ &= (4\sqrt{2})(5) + (4\sqrt{2})(\sqrt{6}) \\ &= 20\sqrt{2} + (4\sqrt{2 \cdot 6}) \\ &= 20\sqrt{2} + (4\sqrt{12}) \\ &= 20\sqrt{2} + (4 \cdot 2\sqrt{3}) \\ &= 20\sqrt{2} + 8\sqrt{3}\end{aligned}$$

The area is $20\sqrt{2} + 8\sqrt{3}$ units². ■

Many formulas also include roots or powers. For example, the formula for electrical power is $P = \frac{V^2}{R}$, where P represents power (watts), R represents resistance (ohms), and V represents voltage.

EXAMPLE 3

What is the voltage used by a radio rated at 100 watts with a resistance of 8 ohms?

SOLUTION

Let $P = 100$ watts and $R = 8$ ohms.

$$P = \frac{V^2}{R}$$
$$100 = \frac{V^2}{8}$$

$$V^2 = (100)(8) = 800 \longrightarrow V = \sqrt{800} = \sqrt{2 \cdot 400} = 20\sqrt{2}$$

The voltage used by the radio is $20\sqrt{2}$ volts. ■

The properties of exponents can be used to transform exponential functions, such as those involving future value of money.

The future-value formula, $F = P(1 + r)^t$, shows how much an investment will be worth after compounding for a given number of years. In the formula, F represents the future value of the investment, P represents the present value of the investment, r represents the rate of growth (the annual percent of interest the investment will earn), and t represents the time in years.

EXAMPLE 4

What is the approximate monthly rate of growth for an investment that has an annual interest rate of 15%?

► **Think About It** Many real-world problems can be modeled by exponential functions that describe how things grow or decay as time passes. Examples include studies of populations, bacteria, radioactive substances, electricity, temperatures, and credit payments.

SOLUTION

The exponential growth portion of the future-value formula is represented by the expression $(1 + r)^t$.

$$(1 + r)^t$$

$$(1 + 0.15)^t \quad \text{Substitute 15\% for } r.$$

$$(1.15)^t \quad \text{Simplify.}$$

The expression for the annual growth is 1.15^t , but you need to model monthly growth. If t is in months, you'd need the base of the monthly growth expression (call the base m) to satisfy the equation $m^{12t} = 1.15^t$. But how do you find that value of m ? The power of a power property can help.

$$m^{12t} = 1.15^t$$

$$m^{12t} = 1.15^{\left(\frac{1}{12} \cdot 12t\right)} \quad \text{Replace } t \text{ with } \frac{1}{12} \cdot 12t.$$

$$m^{12t} = \left(1.15^{\frac{1}{12}}\right)^{12t} \quad \text{Power of Powers Property}$$

So $m = 1.15^{\frac{1}{12}}$, which equals 1.01171491692. To check that this solution works, look at the value of the monthly and annual rate of growth after 24 months and 2 years.

Monthly growth after 24 months	Annual growth after 2 years
$1.01171491692^{24} = 1.3225$	$1.15^2 = 1.3225$

The approximate monthly rate of growth when the annual rate is 15% would be about 1.17%. ■